

APPENDICES

APPENDIX A: Letter Requesting Permission to Conduct the Study and System Development

CAGAYAN DE ORO COLLEGE
PHINMA EDUCATION

To whom it may concern,
Greetings!

We, the undersigned third-year Bachelor of Science in Information Technology students from PHINMA Cagayan de Oro College, namely: **Sean Jonell S. Pablona, Micah Dusil Lago, Shandi Kate Ampo Linaac, Macarambon, Jonaida Jannah Dimakute and Pacaambung Norcaya Mariano**, respectfully write to request your permission to conduct a research study and system development project within your esteemed [Office/Business].

This initiative is in partial fulfillment of the requirements for our course Quantitative Methods leading to Capstone Research 1, which may ultimately lead to the development of a computer-based system tailored to the needs of your organization. As part of our academic undertaking, we seek to conduct field research and system testing within your premises.

We assure you that all information and data collected during the research will be treated with the utmost confidentiality and will be used solely for academic purposes.

We sincerely hope for your favorable consideration and kind support of our proposal. Your approval will greatly contribute to the success of our academic requirements and to our growth as aspiring IT professionals. Thank you very much, and we look forward to your positive response.

Respectfully yours,

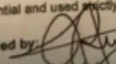
Lago, Micah Dusil Pablona, Sean Jonell Sordilla
Linaac, Shandi Kate Ampo Macarambon, Jonaida Jannah Dimakute
Pacaambung, Norcaya Mariano

BSIT 3rd Year Students
PHINMA Cagayan de Oro College

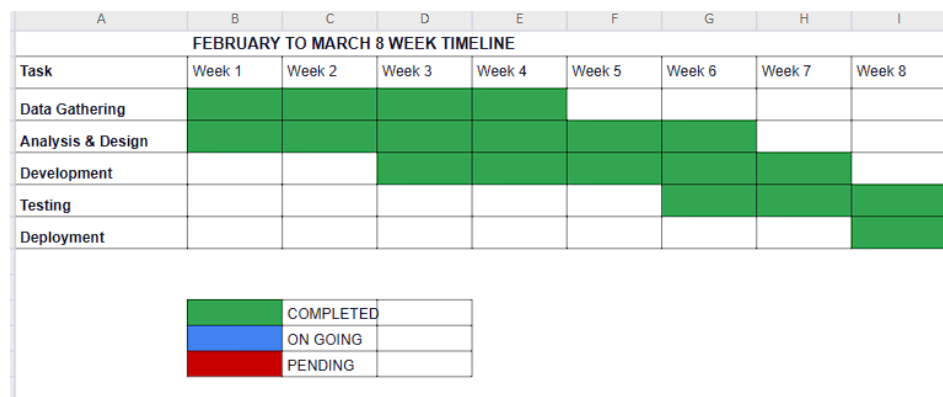
Noted by: 
ALEXIS V. LAROSA, PhD
Instructor/Adviser

Conforme:

I hereby acknowledge and approve the request of the above-mentioned students to conduct their research study and system project within our office/business. I understand that all data gathered will be kept confidential and used strictly for academic purposes.

Approved by: 
RODIN
Name and Signature of Representative

APPENDIX B : Gantt Chart of System Development Activities



APPENDIX C: System Requirements And Requirements Traceability Matrix

C.1 Functional Requirements

The following matrix presents the functional requirements implemented and verified in the developed system.

Requirement ID	Functional Requirement	Description	Verification Method	Status
FR-01	Automated Booking and Scheduling	The system shall allow authorized users to create, manage, and monitor service bookings and schedules while preventing scheduling conflicts.	Automated functional testing and system logs	Verified – 100%
FR-02	Workload-Balanced Personnel Assignment	The system shall assign service tasks based on personnel availability and prevent conflicting assignments.	Conflict-check testing	Verified – 100%
FR-03	Digital Record-Keeping and Document Management	The system shall centrally store and manage operational records and service-related information.	Database and functional testing	Verified – 100%
FR-04	Mobile Synchronization and On-Site Proof Uploads	The mobile application shall allow field personnel to access assigned tasks, update task status, and upload photographic proof.	Mobile API and functional testing	Verified – 100%
FR-05	Structured Income and Transaction Tracking	The system shall record and organize financial and transaction-related information for monitoring and reporting.	Transaction dashboard testing	Verified – 100%
FR-06	Geospatial Service Density Analysis	The system shall represent service transaction locations to	Spatial-coordinate testing	Verified – 100%

		identify geographic service-demand patterns.		
FR-07	Automated Summary Report Generation	The system shall generate operational summaries and reports for management monitoring and decision support.	Reporting dashboard testing	Verified – 100%

C.2 Non-Functional Requirements

Quality Area	Requirement	Implementation/Verification
Security	Restrict access to information according to user roles.	Role-Based Access Control (RBAC)
Data Privacy	Protect personal and operational information.	Data handling aligned with RA 10173
Performance Efficiency	Maintain responsive system operation during use.	Page-load and database-query optimization
Portability	Support use across intended platforms.	Modern web browsers and target Android devices
Reliability	Maintain consistent operation of system functions.	Functional and system testing
Maintainability	Support continued system maintenance and modification.	Structured application and database design

APPENDIX D: System Functionality Testing Documentation

D.1 Unit Testing

Unit testing was conducted to verify individual system functions before integration. The reported results showed that all identified unit tests passed.

Testing Level	Tests Executed	Passed	Failed	Pass Rate
Unit Testing	7	7	0	100%

D.2 Integration Testing

Integration testing was conducted to verify the interaction between connected system modules and services.

Testing Level	Tests Executed	Passed	Failed	Pass Rate
Integration Testing	4	4	0	100%

D.3 System Testing

System testing was conducted to verify the operation of the integrated system.

Testing Level	Tests Executed	Passed	Failed	Pass Rate
System Testing	16*	16*	0	100%

*The asterisk follows the notation used in Chapter 4 and should be reconciled with the final test-case documentation before final submission.

D.4 Functional Test Case Template

Test Case ID	Function Tested	Test Procedure	Expected Result	Actual Result	Status
TC-01	User Authentication	Enter valid login credentials	Authorized user is granted access	As expected	Passed
TC-02	Role-Based Access	Access module according to assigned role	User can access authorized functions only	As expected	Passed
TC-03	Booking Creation	Create a service booking	Booking is recorded successfully	As expected	Passed
TC-04	Schedule Conflict Checking	Create a conflicting schedule	System detects/prevents conflict	As expected	Passed
TC-05	Personnel Assignment	Assign available personnel to a task	Assignment is recorded	As expected	Passed

TC-06	Mobile Task Synchronization	Open assigned task through mobile application	Task information is synchronized	As expected	Passed
TC-07	Status Updating	Change task status	Updated status is reflected in the system	As expected	Passed
TC-08	Proof Photo Upload	Capture/upload service documentation	Image is stored with the relevant task	As expected	Passed
TC-09	Transaction Recording	Record transaction information	Transaction is stored correctly	As expected	Passed
TC-10	Financial Dashboard	Open transaction/financial dashboard	Recorded information is displayed	As expected	Passed
TC-11	Geospatial Analysis	Display transaction locations	Locations are represented spatially	As expected	Passed
TC-12	Report Generation	Generate operational report	System produces the requested summary	As expected	Passed
TC-13	Data Validation	Submit invalid/incomplete information	System prevents invalid submission	As expected	Passed
TC-14	Database Access	Perform authorized database operation	Authorized operation is completed	As expected	Passed
TC-15	Activity Logging	Perform a monitored transaction	System records the activity	As expected	Passed
TC-16	End-to-End Transaction	Complete a service transaction from request to documentation	Workflow is completed successfully	As expected	Passed

APPENDIX E: ISO/IEC 25010 Software Quality Evaluation Questionnaire

Instructions

Please evaluate the developed system based on your actual experience using the system. Place a check mark (✓) beside the response that best represents your assessment.

Rating Scale

Rating	Interpretation
5	Strongly Agree / Excellent
4	Agree / Very Satisfactory
3	Neutral / Satisfactory
2	Disagree / Fair
1	Strongly Disagree / Poor

E.1 Functional Suitability

No.	Statement	5	4	3	2	1
1	The system provides the functions required for service scheduling and transaction management.					
2	The system performs its intended functions accurately.					
3	The system provides appropriate functions for administrative operations.					
4	The system supports field personnel in completing assigned tasks.					
5	The system generates useful operational information and reports.					

E.2 Performance Efficiency

No.	Statement	5	4	3	2	1
1	The system responds promptly to user actions.					
2	The system loads its pages and functions efficiently.					
3	Database transactions are processed efficiently.					
4	Mobile synchronization is performed within an acceptable period.					
5	The system maintains acceptable performance during normal use.					

E.3 Usability

No.	Statement	5	4	3	2	1
1	The system interface is easy to understand.					
2	System functions are easy to locate and use.					
3	The information displayed by the system is understandable.					
4	Users can perform common tasks without excessive assistance.					
5	The system is convenient to use for its intended purpose.					

E.4 Reliability

No.	Statement	5	4	3	2	1
1	The system performs consistently during normal operation.					
2	The system maintains accurate transaction information.					
3	The system correctly maintains user and task information.					
4	The system functions without frequent errors.					
5	The system can be relied upon for routine operational activities.					

E.5 Security

No.	Statement	5	4	3	2	1
1	The system restricts access according to user roles.					
2	User authentication provides appropriate protection.					
3	Operational information is protected from unauthorized access.					
4	The system provides appropriate controls for sensitive information.					
5	System activities can be monitored through appropriate records or logs.					

E.6 Maintainability

No.	Statement	5	4	3	2	1
1	The system structure supports future maintenance.					
2	System modules can be modified without unnecessarily affecting other functions.					
3	Database structures are organized and manageable.					
4	System problems can be identified and investigated.					
5	The system can accommodate reasonable future enhancements.					

E.7 Portability

No.	Statement	5	4	3	2	1
1	The administrative interface can be accessed using modern web browsers.					
2	The system operates properly on the intended devices.					
3	The mobile application functions on the intended Android devices.					
4	System information remains accessible across supported interfaces.					
5	The system can be deployed within its intended operational environment.					

APPENDIX F: ISO/IEC 25010 Evaluation Summary

The following table presents the results of the ISO/IEC 25010 software quality evaluation of the Web-Based Multibranch Music Academy Management System for Father and Sons Music Lessons.

Software Quality Characteristic	Mean	Interpretation
Functional Suitability	5.00	Excellent
Performance Efficiency	4.50	Excellent
Usability & User Experience	4.20	Very Satisfactory
Reliability	3.50	Below Benchmark
Security	5.00	Excellent
Maintainability	4.50	Excellent
Portability	4.50	Excellent
Overall Software Quality	4.47	Excellent

Interpretation: The system achieved an overall mean score of **4.47**, exceeding the 4.0 benchmark and indicating strongly positive evaluator assessment. Five of six characteristics met or exceeded benchmark; Reliability (3.50) was the sole characteristic to fall below it, attributable to a lower fault-tolerance rating (3.0 out of 5.0) for how the system handles erroneous manual data entry.

APPENDIX G: System Usability Scale (Sus) Questionnaire

Instructions

Please indicate how strongly you agree or disagree with each statement based on your experience using the Web-Based Multi-Branch Music Academy Management System for Father and Sons Music Lessons. Put a check mark (✓) in the appropriate column.

Rating Scale

Rating	Meaning
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

SUS Items

No.	Statement	5	4	3	2	1
1	I think that I would like to use this system frequently.					
2	I found the system unnecessarily complex.					
3	I thought the system was easy to use.					
4	I think that I would need the support of a technical person to use this system.					
5	I found the various functions in this system were well integrated.					
6	I thought there was too much inconsistency in this system.					
7	I would imagine that most people would learn to use this system very quickly.					
8	I found the system very cumbersome to use.					
9	I felt very confident using the system.					
10	I needed to learn a lot of things before I could get going with this system.					

SUS Scoring Guide

For positive-numbered items (1, 3, 5, 7, and 9), subtract 1 from the response.

For negative-numbered items (2, 4, 6, 8, and 10), subtract the response from 5.

Add the resulting scores and multiply the total by 2.5 to obtain the SUS score out of 100.

APPENDIX H: System Usability Scale Results

Data from the 10-item System Usability Scale questionnaire were processed using standard SUS scoring rules, where composite scores above 80 represent **Excellent** usability:

- **Calculated System SUS Score:** 84.5 / 100
- **Adjective Rating:** Excellent
- **Acceptability Grade:** Acceptable

Table 4.4: System Usability Scale (SUS) Score Breakdown by Evaluator Group

Evaluator Group Category	Participant Count (N)	Mean SUS Score	Adjective Rating
Administrative Staff	4	85.0	Excellent
Field Technicians	6	82.5	Excellent
IT Experts / Admins	2	86.0	Excellent
General Customers / End-Users	8	84.0	Excellent
Overall Evaluator Mean	20	84.5	Excellent

Interpretation: Achieving an overall SUS score of **84.5 (Excellent)** confirms that administrative operators, technical experts, and mobile field workers found the application interfaces highly intuitive, requiring minimal training or instructional overhead during daily deployment.

The SUS results reported in Chapter 4 indicate that the system received an overall score of 84.5 from the 20 evaluators.

APPENDIX I: Evaluator Group And Functional Testing Matrix

Evaluator/User Group	Primary Functional Scope	Reported Result
IT Experts / Administrators	Authentication, security, RBAC, and system functions	7/7 – 100%
Administrative Staff	Student records, lesson scheduling, attendance, performance, and reporting	4/4 – 100%
Instructors / End-Users	Lesson management, attendance, performance monitoring, and related functions	18/18 – 100%
Total Execution Suite	Unit, Integration, and System Test Cases	29/29 – 100%

APPENDIX J: System Architecture And Design Documentation

J.1 Use Case Diagram

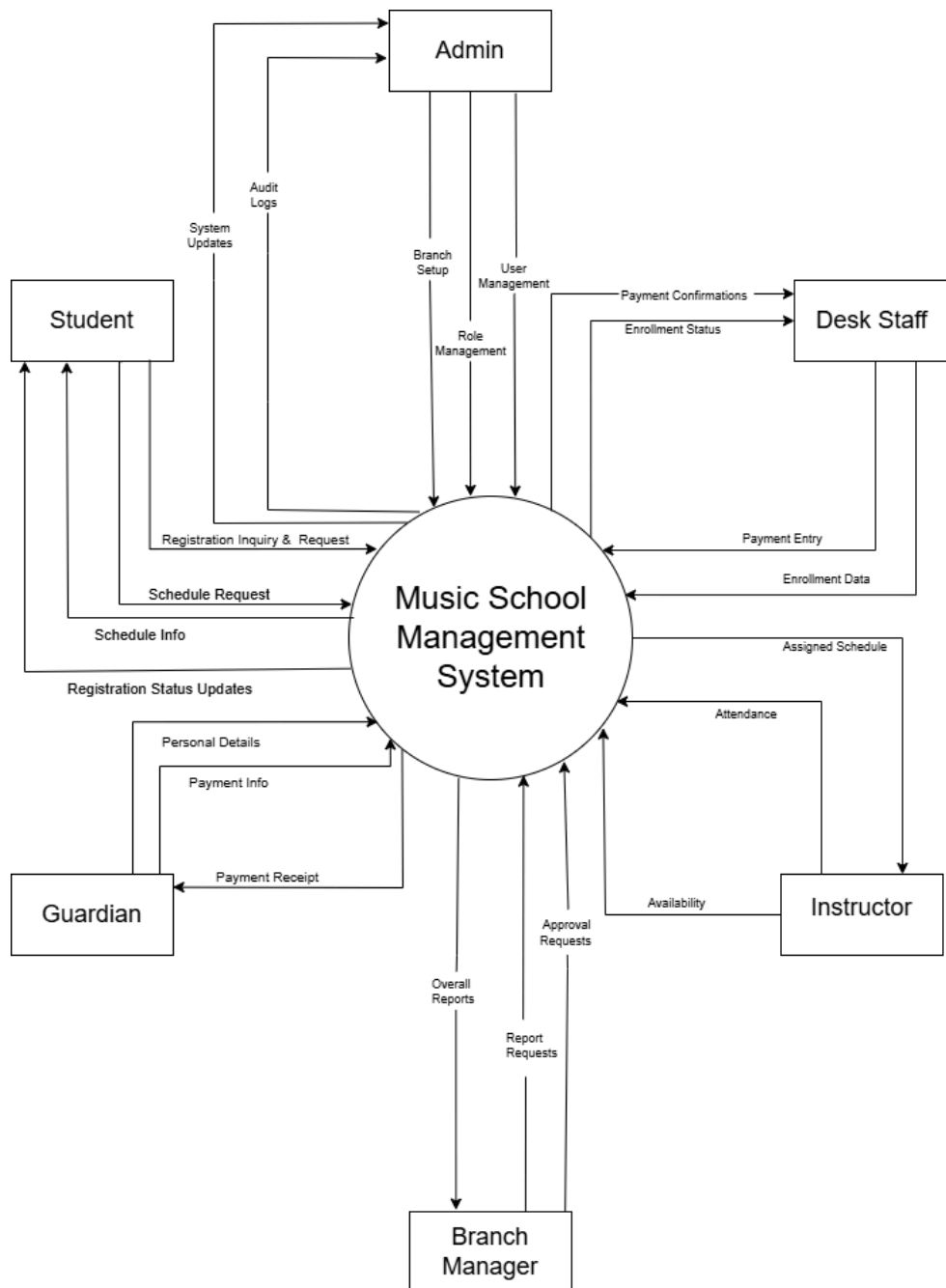
Figure J.1. Use Case Diagram of the Developed Integrated Web-Mobile System



The Use Case Diagram should illustrate the interactions among the principal actors and the system's major functions, including administrative management, scheduling, personnel assignment, transaction management, field-service activities, and status monitoring.

J.2 Context-Level Data Flow Diagram

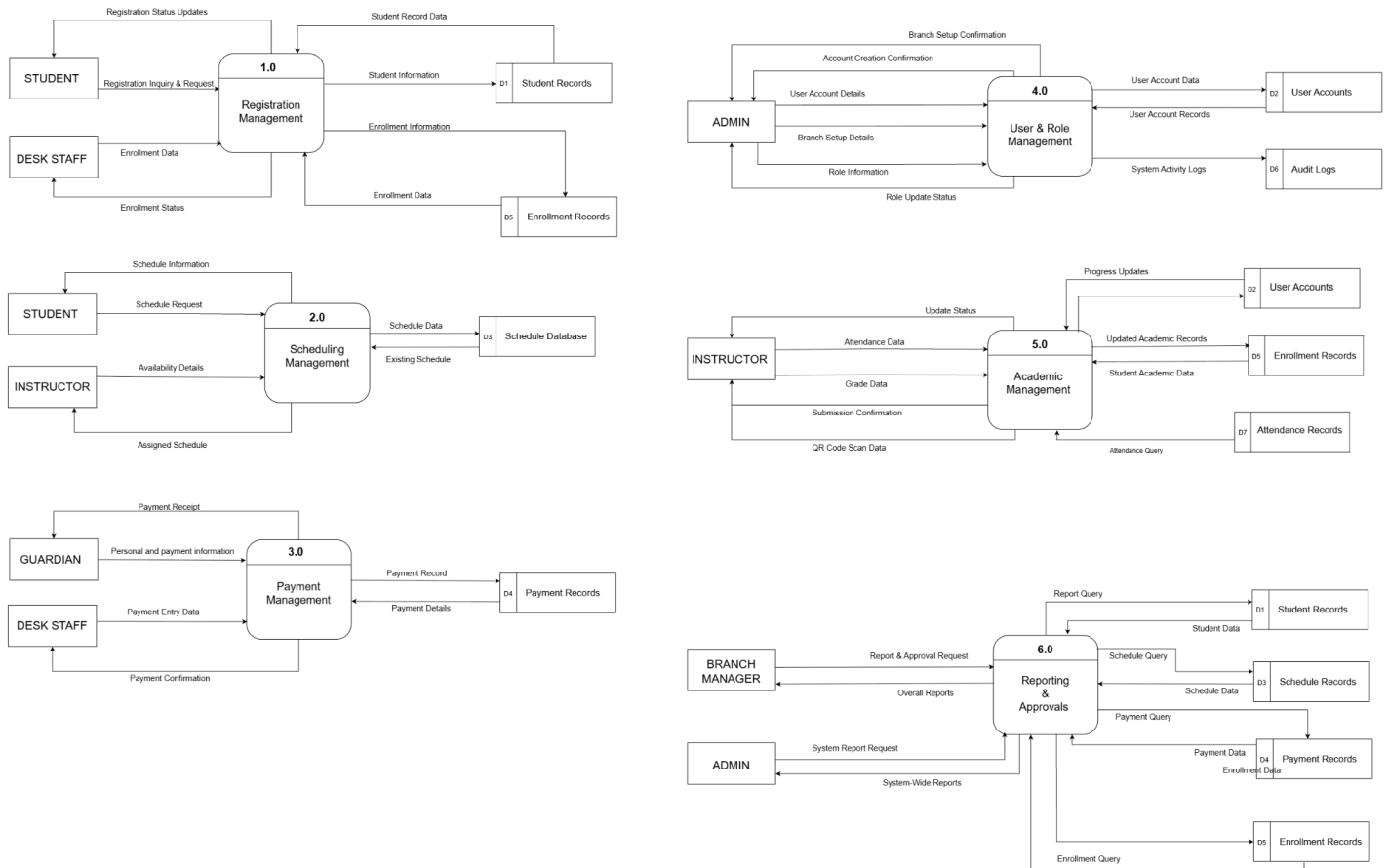
Figure J.2. Context-Level (Level 0) Data Flow Diagram



The diagram should show the developed application as the central processing environment and its interaction with the external actors and operational data sources.

J.3 Level 1 Data Flow Diagram

Figure J.3. Level 1 Data Flow Diagram

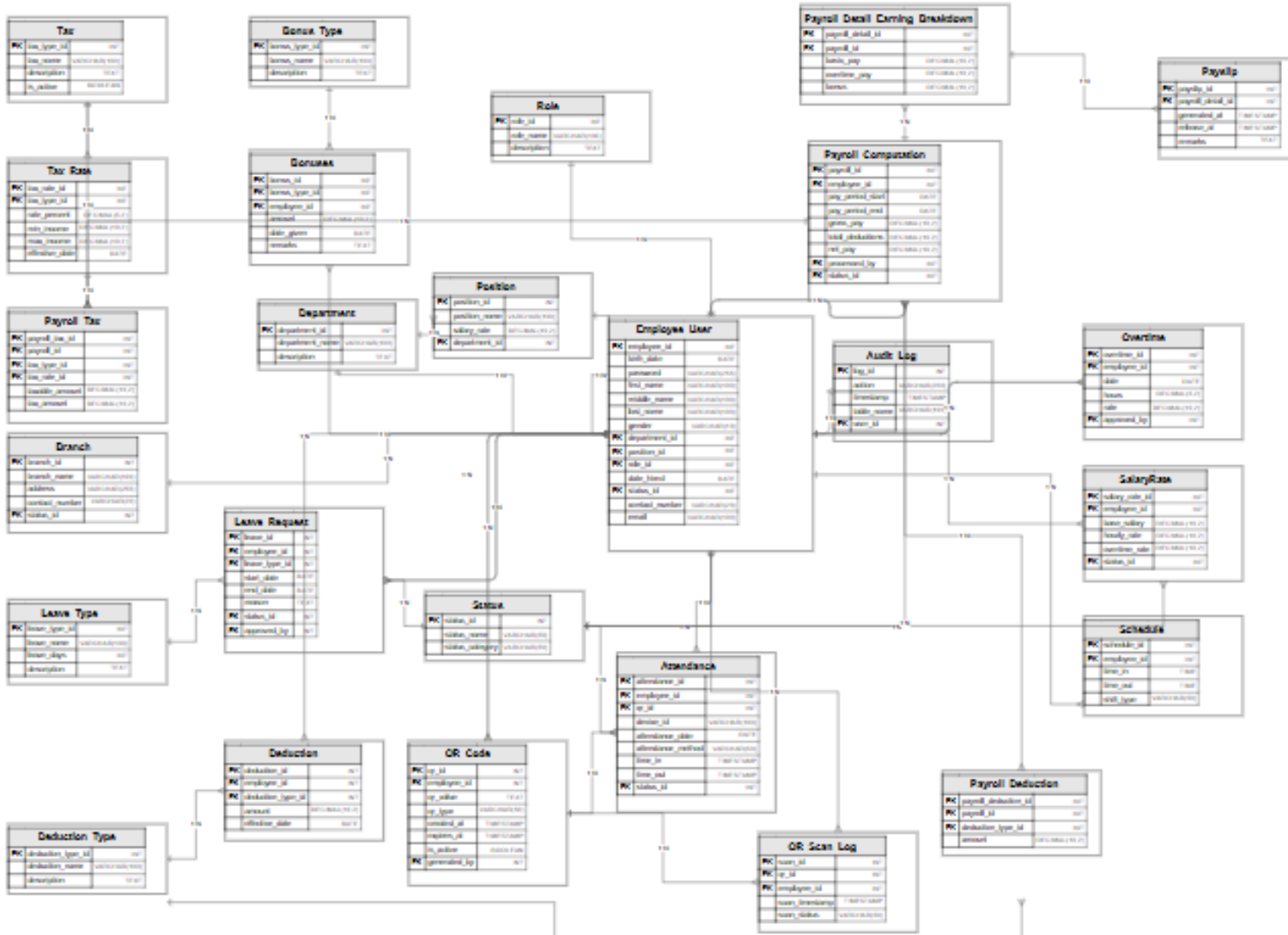


The diagram should illustrate the principal processes, including transaction ingestion, dispatch optimization, and log generation.

J.4 Entity-Relationship Diagram

Figure J.4. Entity-Relationship Diagram

TRAFECO — ENTITY RELATIONSHIP DIAGRAM



The ERD should show the database entities, attributes, primary keys, foreign keys, and relationships used by the developed system.

APPENDIX K: System Interface Documentation

K.1 Administrative Web Portal

K.1.1 Multi-Branch Management Dashboard

Figure K.1. Multi-Branch Management Dashboard

The screenshot displays the 'Registrations' section of the administrative web portal. The left sidebar contains a navigation menu with options like Dashboard, Registrations (highlighted), Students, Users, Instructors, Sessions, Frozen Accounts, Enrollments, Payment Center, Reports, and Audit Logs. The main content area shows a table of student registrations with columns for Student, Guardian, Branch, Fee, Status, Date, and Actions. Four students are listed: Al Christian Abapo, Robin Jay Abas, Yor Kim, and Clove Reyes. Each entry includes a status (Walk-In or Online), a fee of P1000.00, and an 'Approved' status. The 'Actions' column for each student has a 'View' button. The top right of the dashboard shows the user's email (fasadmin@music.com) and a 'Register Student' button.

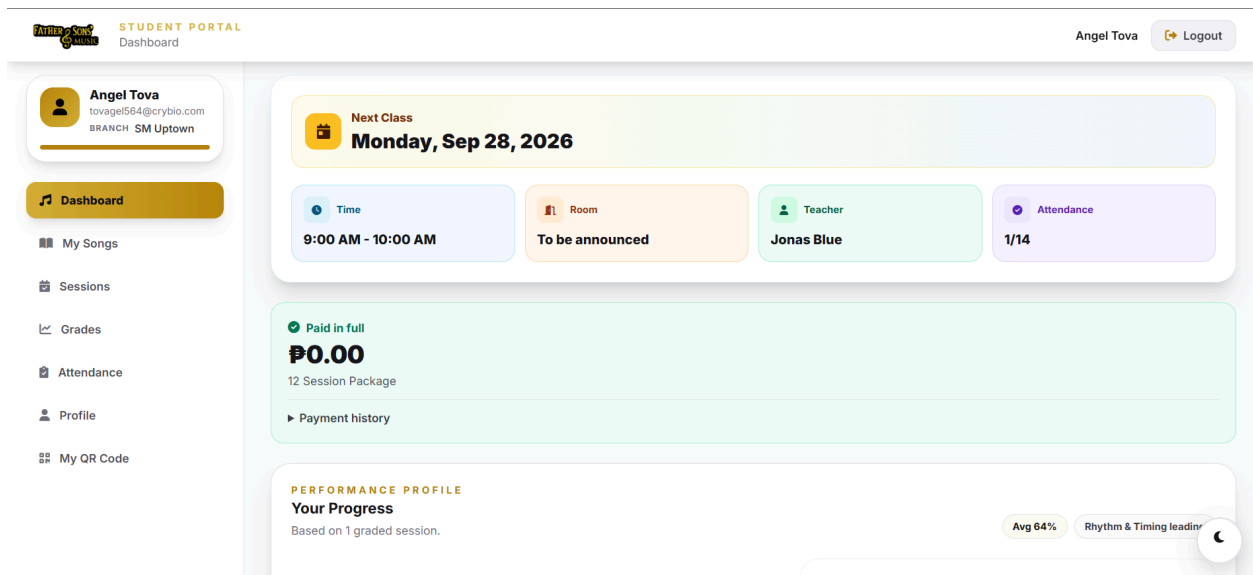
STUDENT	GUARDIAN	BRANCH	FEE	STATUS	DATE	ACTIONS
Al Christian Abapo STU-2026-0191	Walk-In	SM Uptown	P1000.00 Walk-in payment handled at branch Proof ID not required for walk-in	Approved	9/22/2026	View
Robin Jay Abas STU-2026-0190	Online	SM Uptown	P1000.00 Payment ID	Approved	9/22/2026	View
Yor Kim STU-2026-0189	Yuki Kim 0965915309	SM Uptown	P1000.00 Walk-in payment handled at branch Proof ID not required for walk-in	Approved	9/22/2026	View
Clove Reyes STU-2026-0188	Cyan Reyes 0965915309	SM Uptown	P1000.00 Walk-in payment handled at branch	Approved	9/22/2026	View

Description:

The Multi-Branch Management Dashboard provides authorized administrators and branch managers with a centralized overview of Music Academy of Father and Son's Studio. It displays relevant information such as the number of students, instructors, scheduled lessons, attendance records, and branch-related activities for the Uptown and Downtown branches.

K.1.2 Student Management Interface

Figure K.2. Student Management Interface



Description:

The Student Management Interface allows authorized personnel to register, view, update, and manage student information. Student records may include personal information, assigned branch, enrolled lessons, instrument, attendance history, and performance records.

Lesson Scheduling and Management Interface

Figure K.3. Lesson Scheduling and Management Interface

Schedule Student Request
Assign weekly class times based on teacher availability

Student: Ariana Grande Branch: SM Downtown Package: 12 Session Package Instruments: Piano

Start Date
23/09/2026
Classes will repeat weekly from this date

Weekly Time Slots 0 of 1 scheduled

Piano 1
Teacher
GD Dragon

DATE Choose from calendar
TIME No time selected
Pick a highlighted time on the right.

+ Add another day

Choose a date and time from the calendar.

Cancel Assign Schedule

SUN	MON	TUE	WED	THU	FRI	SAT
		1 Unavailable	2 Unavailable	3 Unavailable	4 Unavailable	5 Unavailable
6 Unavailable	7 Occupied	8 Unavailable	9 Occupied	10 Unavailable	11 Unavailable	12 Unavailable
13 Unavailable	14 Occupied	15 Unavailable	16 Unavailable	17 Unavailable	18 Occupied	19 Unavailable
20 Unavailable	21 Occupied	22 Unavailable	23 Available	24 Unavailable	25 Available	26 Unavailable
27 Unavailable	28 Available	29 Unavailable	30 Available			

Description:

The Lesson Scheduling Interface allows authorized users to create, manage, and monitor music lesson schedules. It provides information regarding students, instructors, lesson dates, schedules, instruments, and assigned branches to help minimize scheduling conflicts.

K.1.4 K.4. Instructor Management Interface

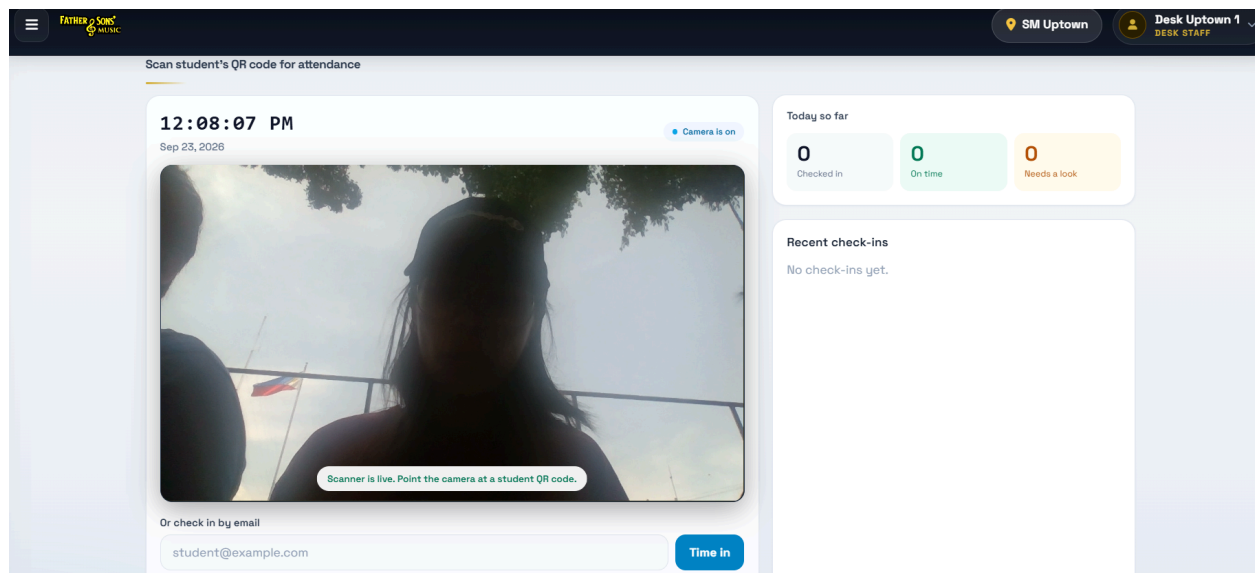
Description:

The Instructor Management Interface allows administrators and branch managers to manage instructor information and assignments. It provides access to instructor profiles, assigned students, lesson schedules, and branch assignments.

K.2 QR Code Attendance Management

K.2.1 Field Task Synchronization

Figure K.5. QR Code Attendance Management Interface



Description:

The QR Code Attendance Management Interface supports digital attendance recording through QR code scanning. The system validates the attendance transaction and records relevant information such as the student, date, time, lesson, and branch.

K.2.2 Attendance Monitoring Dashboard

Figure J.6. Attendance Monitoring Dashboard

Description:

The Attendance Monitoring Dashboard provides administrators, branch managers, and authorized instructors with a consolidated view of student attendance records. It enables users to monitor attendance status, identify attendance patterns, and review attendance history across branches.

K.2.3 Student Performance Monitoring Interface

Figure K.7. Student Performance Monitoring Interface

Description:

The Student Performance Monitoring Interface allows instructors and authorized personnel to record and review student musical progress. Performance records may include lesson performance, skills demonstrated, assessment results, remarks, and progress information related to the student's instrument or enrolled lessons.

Performance Analytics Dashboard

Figure K.8. Performance Analytics Dashboard

Description:

The Performance Analytics Dashboard presents summarized student performance information in an organized format. It assists instructors and administrators in identifying student progress, performance trends, and areas that may require additional attention.

K.1.9 Automated Report Generation Interface

Figure K.9. Automated Report Generation Interface

Description:

The Automated Report Generation Interface allows authorized users to generate consolidated attendance, student performance, lesson, and branch-related reports. The feature reduces the need for manually compiling information from paper records and spreadsheet files.

K.1.10 User and Role Management Interface

Figure K.10. User and Role Management Interface

Description:

The User and Role Management Interface allows authorized administrators to manage system accounts and assign appropriate access privileges. The system supports role-based access for Admin, Branch Manager, Instructor, and Desk Staff to help protect sensitive academy information.

K.2 Student and Instructor Web Interfaces

K.2.1 Student Profile and Lesson Records

Figure K.11. Student Profile and Lesson Records Interface

Description:

The Student Profile and Lesson Records Interface provides access to relevant student information, enrolled lessons, schedules, attendance history, and performance records based on the user's assigned access privileges.

K.2.2 Instructor Lesson Management

Figure K.12. Instructor Lesson Management Interface

Description:

The Instructor Lesson Management Interface allows instructors to view their assigned lessons and students and manage relevant lesson and performance information.

K.2.3 Attendance Recording and Verification

Figure K.13. Attendance Recording and Verification Interface

Description:

The Attendance Recording and Verification Interface displays the result of a QR code attendance transaction and records the student's attendance information in the centralized database.

K.2.4 Student Performance Recording

Figure K.14. Student Performance Recording Interface

Description:

The Student Performance Recording Interface enables authorized instructors to record student progress, assessment results, performance observations, and relevant remarks following music lessons.

The interface features listed above correspond to the system features documented in Chapter 4.

APPENDIX L: System Evaluation Consent And Respondent Information Sheet

Participant Information

You are invited to participate in the evaluation of the developed **Web-Based Multi-Branch Music Academy Management System with Smart QR Code-Based Attendance and Performance Monitoring** for Father and Sons Music Lessons.

The purpose of the evaluation is to determine the functionality, software quality, usability, reliability, security, and operational effectiveness of the developed system in managing music academy activities across the Uptown and Downtown branches.

Participation in the evaluation is voluntary. The information collected from the evaluation shall be used for academic research and system-evaluation purposes.

Confidentiality

Information provided by participants shall be treated confidentially and shall only be used for the purposes of this study. Individual responses shall not be publicly identified in the presentation and analysis of the evaluation results. Collected evaluation data shall be handled according to applicable institutional research procedures.

Evaluation Procedure

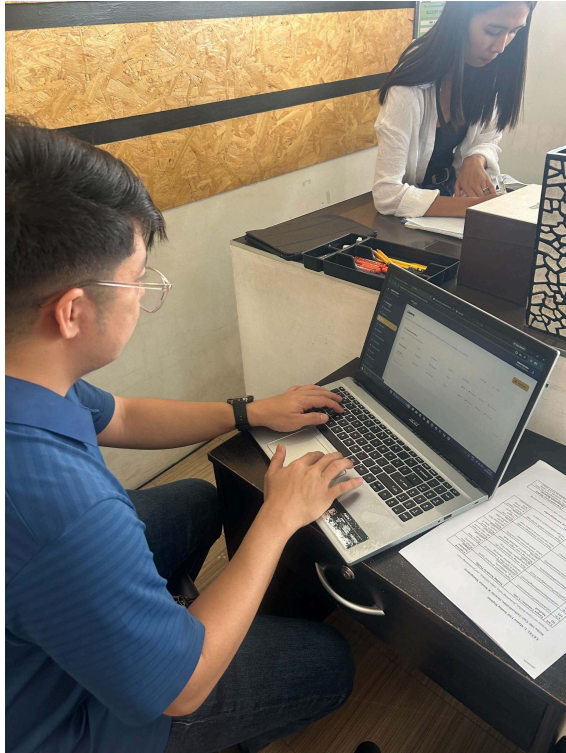
1. Receive an orientation regarding the developed Web-Based Multi-Branch Music Academy Management System;
2. Access or observe the system functions relevant to their assigned role;
3. Perform or observe designated system functions, including student management, lesson scheduling, QR code attendance, and performance monitoring;
4. Test the system according to the assigned evaluation procedures;
5. Complete the ISO/IEC 25010-based evaluation questionnaire, when applicable;
6. Complete the System Usability Scale (SUS) questionnaire, when applicable; and
7. Provide comments, observations, and recommendations regarding the developed system.

Voluntary Participation

Participation in this system evaluation is completely voluntary. Participants may choose whether or not to take part in the evaluation and may skip any question or evaluation item they are not comfortable answering. They may also stop participating at any time without penalty or negative consequences. All participants are encouraged to provide honest feedback based on their experience using or observing the developed system.

APPENDIX M: Documentation Of System Testing And Evaluation

Client Testing Of System



Client Evaluation Meeting



APPENDIX N: Summary Of System Evaluation Results

N.1 Consolidated Results

Evaluation Component	Result
Functional Requirements Verification	100% verified
ISO/IEC 25010 Overall Mean	4.47
ISO/IEC 25010 Interpretation	Strongly Agree / Excellent
Functionality Testing	100% pass rate (29/29 test cases)
Overall SUS Score	84.5 / 100
SUS Adjective Rating	Excellent
SUS Acceptability	Acceptable
Transaction Processing Time	Reduced from 25–30 minutes to approximately 3 minutes
Processing Velocity Improvement	88%
Workflow Steps	Reduced from approximately 8 manual steps to 2 automated triggers
Workflow Complexity Reduction	75%
Information Management	Centralized relational database
Operational Accountability	Automated activity logs

N.2 Overall Evaluation Statement

The overall evaluation results show that the developed Web-Based Multi-Branch Music Academy Management System successfully met the identified functional requirements and received a positive evaluation in terms of software quality. The system obtained an ISO/IEC 25010 overall mean score of 4.47, which was interpreted as Strongly Agree/Excellent. It also achieved a 100% functionality testing pass rate, with all 29 test cases successfully completed, and received an Excellent SUS rating of 84.5 out of 100. Overall, the results indicate that the system helped improve the organization of academy records, provided centralized access to information, and made selected administrative processes more efficient through automation.